Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “system” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Bring back an earlier idea

A 0.4 kg system accelerates at 0.6 m/s².
Predict acceleration for 0.8 kg at the same net force.
Explain the effect of greater resistance.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

Your turn: Draw separate force diagrams for the horizontal interaction.
Explain why equal forces do not imply equal accelerations.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Equal forces, different results

Cart A is 1 kg and cart B is 4 kg. During an ideal horizontal interaction, each force magnitude is 12 N.
Find both acceleration magnitudes and compare them.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

04 Identify actual partners

For a book on a table, name the partner of Earth-on-book and the partner of table-on-book.
Explain why weight and normal are not a third-law pair.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Horse plus cart

The horse pulls cart forward and cart pulls horse backward equally.
Draw a system boundary around both and explain how the system can accelerate forward.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

East is positive. Start at 7 m, go to -5 m, then finish at 4 m.
Sketch the route; calculate distance and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A spacecraft pushes on a massive asteroid.
Compare instantaneous interaction forces and accelerations from the interaction.
State an assumption in your acceleration comparison.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg cart pushes a 3 kg cart on a frictionless track. An external 10 N force acts on the 2 kg cart.
Find their shared acceleration and contact force; draw both diagrams.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3